# Midazolam

****This monograph is under review****

**FORM** Vials containing midazolam 50mg in 50ml

**INDICATION**
1. Sedation
2. Seizure control

**DOSE RANGE**

**1. Sedation**

| AGE | DOSE | FREQUENCY | ROUTE |
|---|---|---|---|
| Neonate – 6months | 100micrograms/kg | Single dose | I.V. |
| | 30 - 60micrograms/kg/hour* | Continuous Infusion | I.V. |

* Lower maintenance doses may be required in preterm neonates and any patient already receiving opiates. Adjust dose according to response.

**2. Seizure control**

| AGE | DOSE | FREQUENCY | ROUTE |
|---|---|---|---|
| Neonate – 6months | 150 micrograms /kg | Single loading dose | I.V. |
| | Then 60micrograms/kg/hour<br>Increase dose by 60micrograms/kg/hour every 15minutes until seizures controlled<br>**Maximum 300micrograms/kg/hour** | Continuous Infusion | I.V. |

Caution in renal, liver impairment and hypothermia as accumulation may occur. A reduced dose may be required

**PRESCRIPTION OF CONTINUOUS INFUSION**

**SINGLE STRENGTH = 6mg/kg in 50ml infusion fluid**
This gives:-
- 60micrograms/kg/hour at 0.5ml/hr
- 120micrograms/kg/hour at 1ml/hr

**DOUBLE STRENGTH = 12mg/kg in 50ml infusion fluid**
This gives:-
- 60micrograms/kg/hour at 0.25ml/hr
- 120micrograms/kg/hour at 0.5ml/hr

**RECONSTITUTION** Already in solution

**DILUTION**

**For continuous Infusion**

**SINGLE STRENGTH**
**Using 50mg/50ml (1mg/ml) injection**
6 x wt (kg) is the number of ml of 50mg/50ml to be diluted up to 50ml total with infusion fluid (equivalent to **6mg/kg in 50ml**)

**DOUBLE STRENGTH**
**Using 50mg/50ml (1mg/ml) injection**
12 x wt (kg) is the number of ml of 50mg/50ml to be diluted up to 50ml total with infusion fluid (equivalent to **12mg/kg in 50ml**)

**For slow IV injection**

| | |
|---|---|
| Midazolam 1mg/ml inj. | 1ml |

| | |
|---|---|
| 0.9% Sodium Chloride Inj. | Up to 10ml total |

Gives a 100 microgram in 1ml solution. Use the required volume.

**METHOD OF ADMINISTRATION**

**For continuous Infusion**
By continuous intravenous infusion, flow rate adjusted according to the baby's response (see prescription section for details).

**For slow IV injection**
By slow IV injection over 5 minutes.
**Note rapid IV injection (less than 2minutes) may cause seizure-like myoclonus in preterm neonates.**

**Slow IV injection while on continuous infusion**
As above for slow IV injection **OR** given from the continuous infusion syringe (see above) as:
0.8ml over 5 minutes(100micrograms/kg) - **SINGLE strength**
Or
0.4ml over 5 minutes (100micrograms/kg) - **DOUBLE strength**

**Hands free bolus from infusion**
**SINGLE strength infusion** - 10ml/hr over 5 minutes= 100mcg/kg

**COMPATIBILITY**

| | |
|---|---|
| **Solution compatibility** | sodium chloride 0.45%, sodium chloride 0.9%, Glucose 5%, Glucose 10% |
| **Solution incompatibility** | No Information |
| **IV Line compatibility** | Adrenaline, Dobutamine, Dopamine, Heparin, Insulin, Metronidazole, Morphine, Ranitidine, Vancomycin, Vecuronium |
| **IV Line incompatibility** | Aciclovir, Amphotericin, Dexamethasone, Furosemide, Hydrocortisone, Sodium bicarbonate, Phenytoin, Phenobarbital, TPN |

**THIS LIST IS NOT EXHAUSTIVE PLEASE CONTACT PHARMACY FOR FURTHER INFORMATION ON COMPATIBILITY WITH ANY MEDICINES NOT INCLUDED**

**CAUTIONS, CONTRA-INDICATIONS AND SIDE EFFECTS**
See Summary of Product Characteristics and most recent edition of BNF for Children

**FURTHER INFORMATION**

- Time to onset is 2-3 minutes.
- **Flumazenil** is the specific antidote for midazolam induced respiratory depression.
- Erythromycin, other macrolides and possibly fluconazole inhibit the metabolism of midazolam, resulting in profound sedation.
- Injection is painful and may cause thrombophlebitis.

**PH** 2.9-3.7

**LICENSED STATUS** Unlicensed in children <6months old

**APPLICABLE POLICIES** West of Scotland Neonatal Guidelines
Consult local policy if applicable

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|---|---|---|---|
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**Administer reconstituted solutions immediately.**
**All vials, ampoules and infusion bags are for single use only unless otherwise stated.**

Dose may vary depending on indication, age, renal function, hepatic function, and concomitant medications.
This monograph should be used in conjunction with the terms of reference document prepared by the West of Scotland Pharmacist Network. Information is correct at the time of publication and as per local practice agreement.
For further advice please contact your clinical pharmacist or pharmacy department